# RAG Assignment: Baseline vs. Re-Ranking

Analysis and Report

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October 15, 2026

## Pipeline Setup

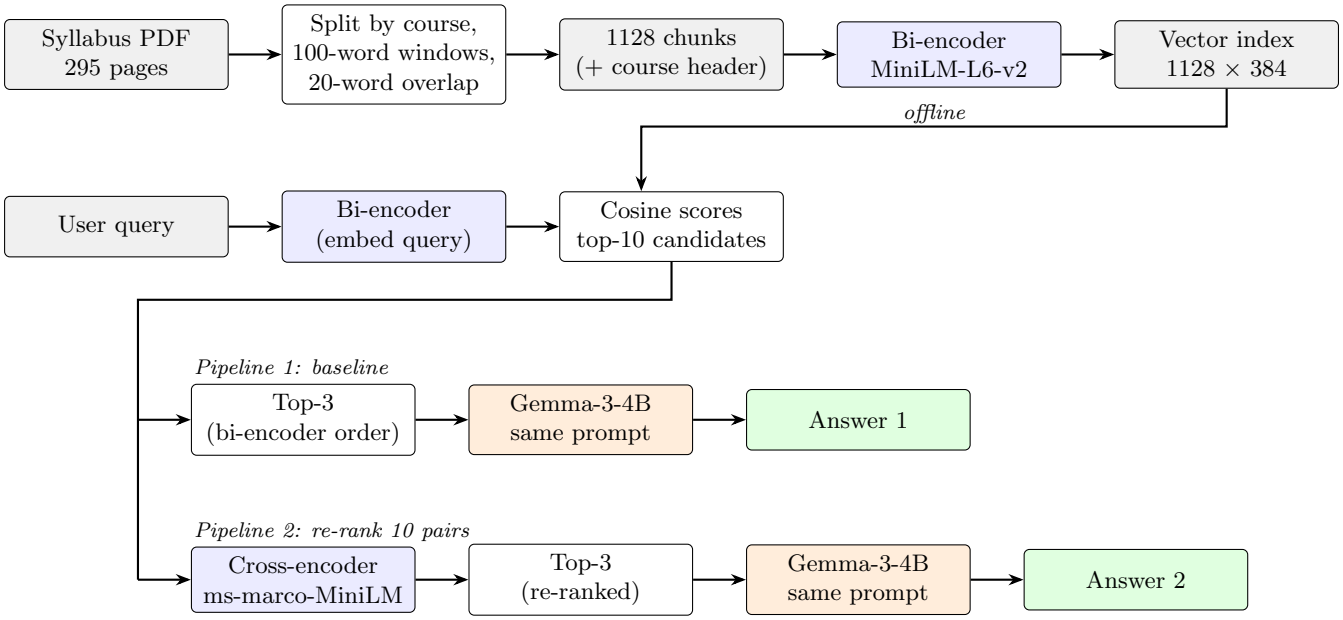



Figure 1: Setup

# 1 Architecture Comparison

# 2 Architecture Comparison

- **Bi-encoder:** encodes the query $q$ and chunk $d$ *separately* into vectors $\mathbf{q}$, $\mathbf{d}$, compared by

$$\text{sim}(q, d) = \frac{\mathbf{q} \cdot \mathbf{d}}{\|\mathbf{q}\| \, \|\mathbf{d}\|}.$$

- Chunk vectors are precomputed once, so search is fast.
- **Cross-encoder:** reads $(q, d)$ together as one input. Joint self-attention across all tokens gives one relevance score .
- More precise, but nothing can be precomputed.One forward pass per (query, chunk) pair at query time.

## 3 Qualitative and Rank Shift Analysis

I tested 31 queries covering keyword traps, exceptions, negation and buried facts. For each one I marked the gold chunk using an exact phrase from the PDF. Out of the 31, 22 were already answered correctly by the baseline, 2 were fixed by re-ranking, 7 were recall failures where the right chunk was not in the top 10, and none got worse after re-ranking.

| Query | Rank of gold chunk (bi-encoder to cross-encoder) | Answer 1 (baseline) | Answer 2 (re-ranked) |
|---|---|---|---|
| Which course uses an open book exam followed by a project with flight testing? | UAV Design chunk (page 153): rank 4 to rank 1 | "The provided passages do not contain the answer." | Correct, cited [1]: "Open book exam followed by detailed project submission including simulation studies, prototype development and flight testing." |
| Is there a limit on how many students can register for Autonomous Software Engineering? | Chunk on page 31: rank 5 to rank 1 | Refused | Also refused |
| Which course has an L-T-P-C of 3-0-2-6? (recall failure) | Gold chunk at rank 424, never in the top 10 | Refused | Refused |

Table 1: Rank changes and answers for the three analysed queries.

**Open book exam with flight testing.** The query was "Which course uses an open book exam followed by a project with flight testing?"

- The right chunk, from Introduction to UAV Design on page 153, moved from rank 4 to rank 1 after re-ranking.
- The baseline top 3 came from Digital VLSI Design, Wireless Communications and Probability and Statistics. They talk about group projects, so they look similar to the question, but they are the wrong courses.
- Answer 1 was "The provided passages do not contain the answer."
- Answer 2 quoted the correct sentence: "Open book exam followed by detailed project submission including simulation studies, prototype development and flight testing."
- The right chunk was outside the top 3 for the baseline, so the model had no evidence. Re-ranking fixed this.

**Intake limit of Autonomous Software Engineering.** The query was "Is there a limit on how many students can register for Autonomous Software Engineering?"

- The right chunk, on page 31 with the line "In Take Limit : 30", moved from rank 5 to rank 1 after re-ranking.
- The baseline top 3 were all from the right course, but none of them had the intake line.
- Both Answer 1 and Answer 2 were "The provided passages do not contain the answer."

- So re-ranking fixed the retrieval but not the answer. The chunk only has the bare field "In Take Limit : 30", while the question asks how many students can register.
- When I asked "What is the In Take Limit of Autonomous Software Engineering?", the model answered "The In Take Limit of Autonomous Software Engineering is 30 [1]." So the problem was the difference in wording and the strict refusal rule in the prompt, not retrieval.

**Recall failures.**

- For some queries the right chunk was far outside the top 10. For the L-T-P-C query it was at rank 424, so both pipelines refused.
- A re-ranker can only reorder the candidates it is given, so it cannot fix this.
- Exact tokens like 3-0-2-6 are also hard for a semantic embedding to match.

**Without retrieval.**

- When I asked the model the same questions with no passages, it answered confidently but wrongly, for example by inventing Udacity details and naming "Aviation Meteorology".
- This is why the model needs retrieval for a document it has never seen.

## 4 Latency Profiling

I ran each query 10 times on a T4 GPU, skipped the slow first runs, made sure the clock only stopped after the GPU finished, and averaged everything over 31 queries on a collection of 1128 chunks.

| Stage | Mean time (ms) |
|---|---|
| Pipeline 1: embed query + score all chunks + top-10 | 9.6 |
| Re-ranking step only: cross-encoder on 10 pairs | 31.9 |
| Pipeline 2: baseline retrieval + re-ranking | 43.2 |
| LLM answer generation (same for both) | 2714.4 |

Table 2: Latency per query.

Re-ranking makes retrieval about $43.2/9.6 \approx 4.5\times$ slower, but this adds only about 34 ms, small next to roughly 2.7 s of LLM generation. (The LLM time was measured on 3 queries, so it is approximate.)

## 5 Scaling Question

The bi-encoder prepares the vector of every chunk once, in advance. When a question comes, it only converts the question into a vector and compares it with the saved ones, which is fast.

The cross-encoder cannot prepare anything in advance, because its score depends on the question and the chunk together. So for every question it has to read every chunk again from scratch. Its time therefore grows with the number of chunks $N$:

$$T_{\text{cross}} \approx N \cdot t_{\text{pair}},$$

where $t_{\text{pair}} \approx 2.82\,\text{ms}$ was measured in my run.

| Approach | Time per query |
|---|---|
| Bi-encoder over 1128 chunks | 9.6 ms |
| Cross-encoder on top 10 only | 31.9 ms |
| Cross-encoder on all 1128 chunks | 3.18 s |
| Cross-encoder on $10^6$ chunks (extrapolated) | $\approx$ 47 min |

Table 3: Cost of cross-encoding the whole collection. The last row is a straight-line estimate from the T4 measurement.

Even on this small collection, running the cross-encoder on everything is about $3180.6/9.6 \approx 331$ times slower than the bi-encoder. For a million chunks, one question would take about 47 minutes, which is not usable.